

ANALYZE *Amazon* REVIEW

Review Analysis and Prediction



Kelompok 2



01

PEMBAGIAN TUGAS

01 MAJID

- Insight Finding
- Experiment : Baseline Undersampling dan Natural Language Toolkit

02 RIO

- Anomaly Detection
- Experiment: Weighted class, Undersampleing, Ekstraksi N-Grams, SMOTE

03 NEVA

- Text Preprocessing
- Feature Engineering/Extraction
- Experiment: Baseline, Ensemble Training , Sampling

04 ARSYAD

- Anomaly Handling
- Experiment: Weighted class, hyperparameter tuning, sampling

02 DATASET OVERVIEW

DATA SHAPE



Jumlah Baris: 34660

Jumlah Kolom: 21

FITUR NUMERIKAL



FITUR KATEGORIKAL

- 'reviews.id'
- 'reviews.numHelpful'
- 'reviews.rating'
- 'reviews.userCity'
- 'reviews.userProvince'

- | | |
|------------------|-------------------------|
| • 'id' | • 'reviews.dateAdded' |
| • 'name' | • 'reviews.dateSeen' |
| • 'asins' | • 'reviews.didPurchase' |
| • 'brand' | • 'reviews.doRecommend' |
| • 'categories' | • 'reviews.sourceURLs' |
| • 'keys' | • 'reviews.text' |
| • 'manufacturer' | • 'reviews.title' |
| • 'reviews.date' | • 'reviews.username' |

1

Exploratory Data Analysis

insight finding

- Distribusi Rating
- Produk & Brand Terpopuler
- Rata-Rata Rating per Brand
- Tren Review per Bulan
- Panjang Teks vs Rating
- Review Paling Membantu
- Imbalance Check
- Proporsi doRecommend
- Korelasi Fitur Numerik

Anomaly finding

- Missing Value (16 kolom)
- Mix Data Type (float, str)
- Outlier (fitur numHelpful : 814)
- Case Inconsistency (Amazon | AMAZON)
- Punctuation Noise (',!@\$%?')
- Typo and Slang Vocabulary Noise (wemo, k8d, couldn)

2

Data Cleaning

NaN Handling

- Drop kolom ga penting
- Drop kolom noise (Nan > 50%)
- Isi kolom panjang text

Case Inconsistency

- Lower case dan strip

Punctuation noise

- Hapus semua tanda baca & symbol

Undersampling

- Kelas positif
- Random
- Stratify y
- 32k → 900

3

Feature Engineering

Fitur Independen

- VADER Sentiment
- POS Tagging
- Flesch Reading Ease
- Words Count
- Text Length

Fitur Dependen

- TF-IDF
- Bag-of-Words
- LDA Topic Features

4 Modeling

Train Test Split

- Train: 80
- Test: 20

Class Weightening

- Negatif: 1.32
- Netral: 0.71
- Positif: 1.18

Hyperparameter tunning

- Optuna
- Cross val: 5 fold
- Scoring: f1_macro

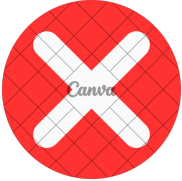
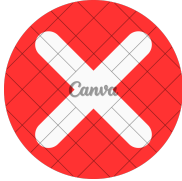
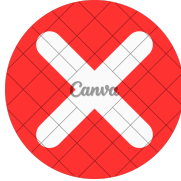
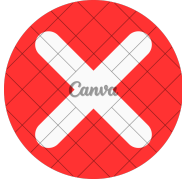
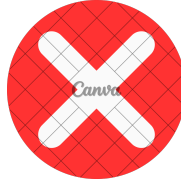
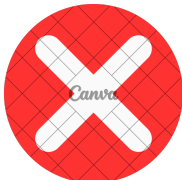
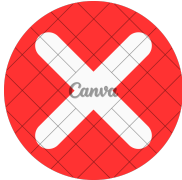
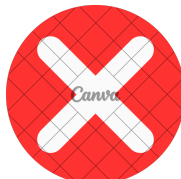
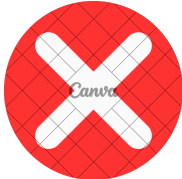
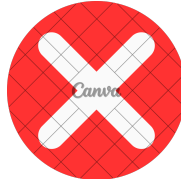
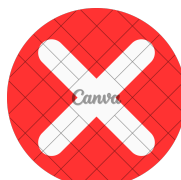
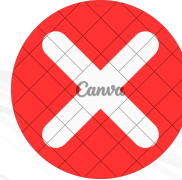
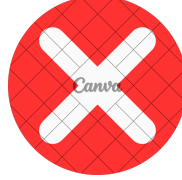
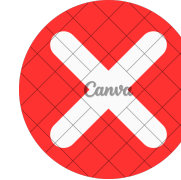
5 Evaluasi

Metrik

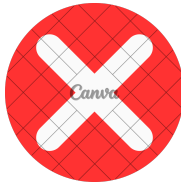
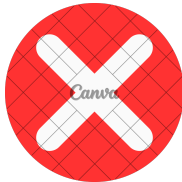
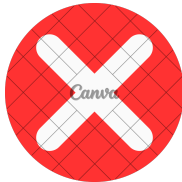
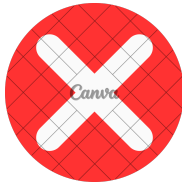
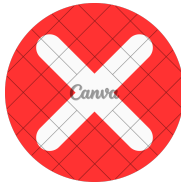
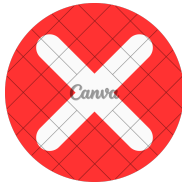
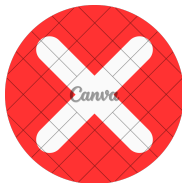
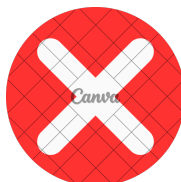
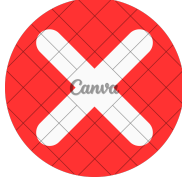
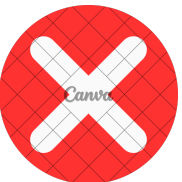
- F1 Macro
- Confusion Matrix

05

EXPERIMENT RESULTS

Model	Undersamplin g	Oversampling	Class Weightening	N-Gram	Hyperparame ter tuning	F1-Macro
Naive Bayes						32%
Naive Bayes						65%
Naive Bayes						punya neva
Light GBM						53%
Naive Bayes						43%

07 **EXPERIMENT RESULTS**

Model	Undersamplin g	Oversampling	Class Weightening	N-Gram	Hyperparame ter tuning	F1-Macro
Random Forest						34%
Random Forest						70%
SVM						49%
Ensemble (XGB + LGBM)						58%

08 BEST MODEL: RANDOM FOREST

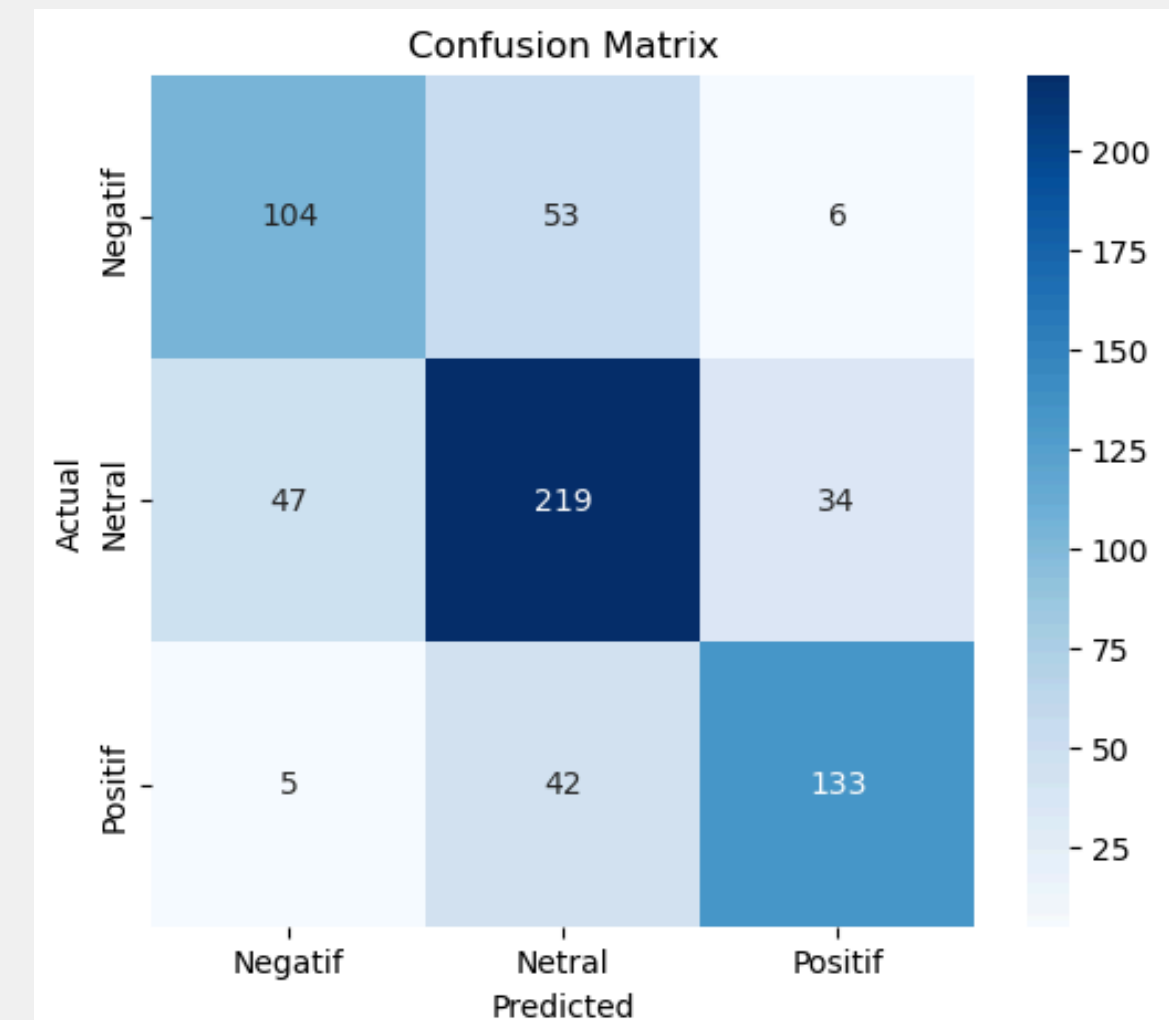
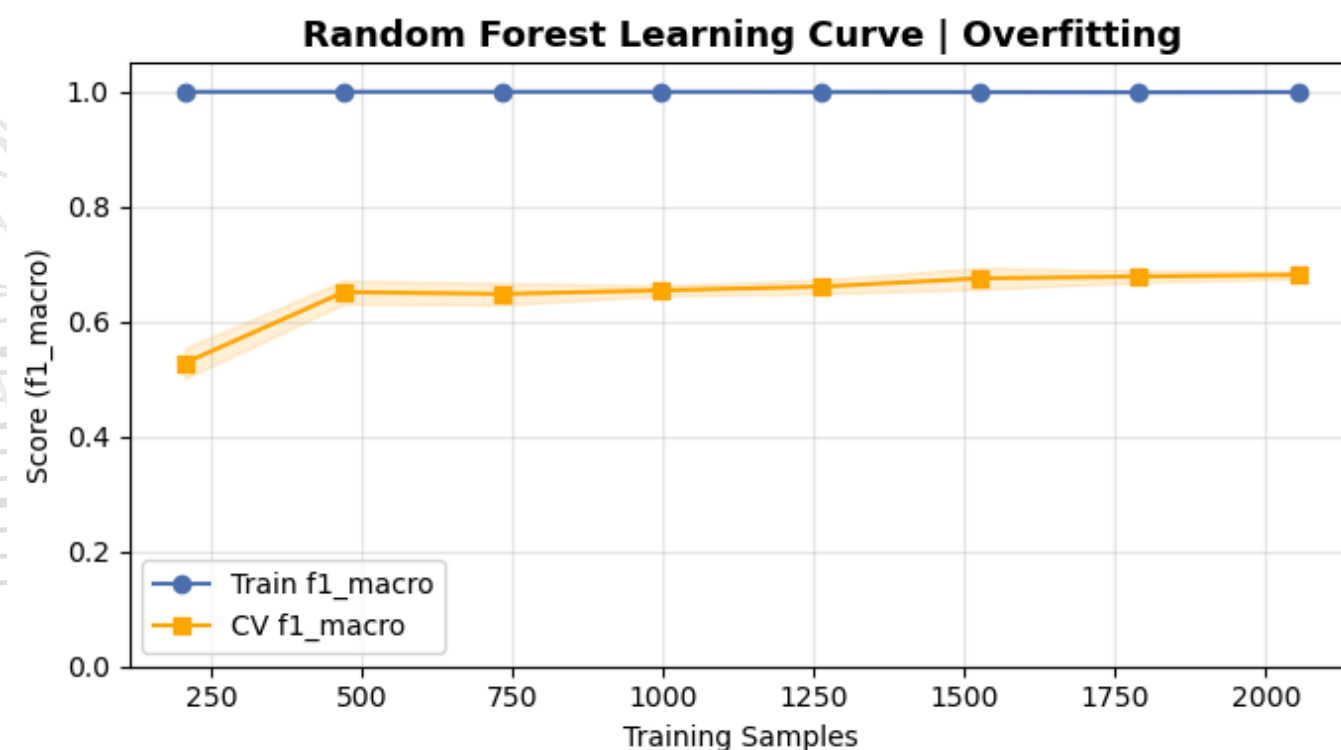
BEST PARAMETER:



n_estimators: 333,
'max_depth': 43,
'max_features': 'sqrt',
'min_samples_split': 5,
'min_samples_leaf': 1,
'bootstrap': False

PERLAKUAN EXPERIMENT:

Undersampling: 32k → 900
Weighted Class
Hyperparameter tuning



label	precision	recall	f1-score
Negatif	67%	64%	65%
Netral	70%	71%	71%
Positif	77%	75%	75%

Train Acc : 99%
Test Acc : 70%

Test Precision : 70%
Test Recall : 70%

Test F1 (macro) : 70%